

Graphs of functions - Set 2 - Solutions

Specification MM8.1, MM8.5-MM8.7, M4.15.

Answer key

Question	Answer	Topic
1	A	Graphs of functions
2	A	Graphs of functions
3	E	Graphs of functions
4	F	Graphs of functions
5	B	Graphs of functions
6	G	Graphs of functions

Worked solutions

Question 1 GRA-11 · Graphs of functions · spec MM8.7

The curves $y = 2^x$ and $y = x^2$ intersect at how many points?

Answer: A

They meet at $x = 2$ and $x = 4$ ($4 = 4$, $16 = 16$), and once for negative x (at $x = 0$: $1 > 0$; at $x = -1$: $\frac{1}{2} < 1$). For $x > 4$, 2^x grows faster. Three intersections.

Full method - how and why

Exponential versus quadratic. Test the obvious integer points and use end behaviour to be sure nothing is missed.

- 1 Try small integers: $x = 2$ gives $4 = 4$ ✓ and $x = 4$ gives $16 = 16$ ✓ - two intersections already.
why: $2^x = x^2$ has these two well-known solutions.
- 2 Check negative x : at $x = 0$, $2^0 = 1 > 0 = x^2$; at $x = -1$, $2^{-1} = \frac{1}{2} < 1 = x^2$. The order swaps, so there is a crossing between -1 and 0 .
why: A sign change in the difference $2^x - x^2$ guarantees a crossing.
- 3 Check beyond $x = 4$: 2^x grows faster than x^2 (e.g. at $x = 5$, $32 > 25$), so they never meet again.
why: Exponential growth eventually outpaces any polynomial.
- 4 Total: three intersections.
why: One negative, and $x = 2$, $x = 4$.

Where it goes wrong

- Finding only $x = 2$ and $x = 4$ and answering 2.
- Forgetting that x^2 is positive for negative x while 2^x stays small but positive.

Takeaway Test integer points, then check the two ends: a sign change in the difference means an extra crossing.

Question 2 GRA-13 · Graphs of functions · spec MM8.7

For how many values of k does the equation $|x^2 - 4| = k$ have exactly three real solutions?

Answer: A

The graph of $y = |x^2 - 4|$ is a W-shape with minima 0 at $x = \pm 2$ and a local maximum 4 at $x = 0$. A horizontal line gives three solutions only when it passes through the local maximum: $k = 4$. One value.

Full method - how and why

Sliding a horizontal line across a modulus graph. Three solutions happen only at the height of the local maximum.

- 1 Sketch $y = |x^2 - 4|$: the parabola $x^2 - 4$ with the part between $x = -2$ and $x = 2$ flipped above the axis.
why: Reflecting the negative part gives a W-shape with two zeros and a hump.
- 2 Identify the key heights: zeros at $x = \pm 2$, and a local maximum of 4 at $x = 0$.
why: At $x = 0$, $|0 - 4| = 4$ - the top of the central hump.
- 3 Count intersections with $y = k$: for $0 < k < 4$ there are four; for $k > 4$ there are two; for $k = 0$ there are two; for $k = 4$ the line passes through the hump's peak **and** cuts the outer arms: three.
why: At the peak, two of the four intersections merge into one.
- 4 So exactly one value of k (namely 4) gives three solutions.
why: The question asks how many values of k , not how many solutions.

Where it goes wrong

- Answering 4 (the value of k) instead of 1 (the number of such k).
- Forgetting $k = 0$ gives two solutions, not three.

Takeaway Slide the horizontal line and note the heights where the count changes: those are the turning values.

Question 3 GRA-14 · Graphs of functions · spec MM8.1

The graph of $y = \sqrt{x}$ is transformed to $y = \sqrt{4 - x}$. Which sequence of transformations achieves this?

Answer: E

$\sqrt{4 - x} = \sqrt{-(x - 4)}$: reflect $y = \sqrt{x}$ in the y -axis to get $\sqrt{-x}$, then translate 4 to the right. (Translating first then reflecting gives $\sqrt{-x - 4}$.)

Full method - how and why

Order matters when combining a reflection with a translation. Rewrite the target to reveal the correct sequence.

- 1 Rewrite the target: $\sqrt{4 - x} = \sqrt{-(x - 4)}$.
why: Factoring out the -1 exposes both operations: a negation of x and a shift.
- 2 Reading it as $f(-(x - 4))$ where $f(x) = \sqrt{x}$: first reflect in the y -axis (giving $\sqrt{-x}$), then translate 4 right (replacing x by $x - 4$).
why: For nested transformations, the operation applied directly to x (here the shift) is done **last**.
- 3 Check with a point: $y = \sqrt{4 - x}$ passes through $(4, 0)$; $\sqrt{-x}$ passes through $(0, 0)$, which moves to $(4, 0)$ after shifting right ✓.
why: A point check settles order questions decisively.
- 4 Verify the other order fails: translating first gives $\sqrt{x - 4}$, then reflecting gives $\sqrt{-x - 4}$ - not the target.
why: Different sequences give genuinely different graphs.

Where it goes wrong

- Choosing 'translate then reflect', which gives $\sqrt{-x - 4}$.
- Reflecting in the x -axis instead of the y -axis.

Takeaway Rewrite as $f(\pm(x - h))$ and test a point; with mixed reflections and shifts the order is not interchangeable.

Question 4 GRA-15 · Graphs of functions · spec MM8.6

The cubic $y = x^3 + ax + b$ has three distinct real roots. Which of the following statements must be true?

- I $a < 0$
- II $b \neq 0$
- III The curve has two turning points.

Answer: F

Three distinct real roots require two turning points (III), which needs $y' = 3x^2 + a$ to have two roots, so $a < 0$ (I). II is false: $x^3 - x$ has roots $0, \pm 1$ and $b = 0$.

Full method - how and why

Conditions for a cubic to have three distinct roots. Two claims follow from the shape; the third needs a counterexample.

- 1 III: three distinct real roots force the curve to turn twice (up, down, up), so it must have two turning points.

True.

why: Between consecutive roots the curve must turn around.

- 2 I: $y' = 3x^2 + a$ has two real roots only when $a < 0$ (so that $x^2 = -\frac{a}{3}$ has solutions). **True.**

why: This follows directly from III: two turning points require y' to have two roots.

- 3 II: test $y = x^3 - x = x(x - 1)(x + 1)$: three distinct roots $0, \pm 1$, and here $b = 0$. **False.**

why: One counterexample is enough; $b = 0$ simply means the origin is a root.

- 4 I and III only.

why: Match to the options.

Where it goes wrong

- Assuming $b \neq 0$ because 'otherwise the cubic factorises' - it does, and that is fine.
- Failing to link I to III and guessing the sign of a .

Takeaway Three distinct roots $\Leftrightarrow$ two turning points with values of opposite sign; that forces $a < 0$ but says nothing about b .

Question 5 X2-09 · Graphs of functions · spec MM8.7

The graphs of $y = f(x)$ and $y = g(x)$ meet at exactly three points. Which of the following statements must be true?

- I The equation $f(x) = g(x)$ has exactly three real solutions.
- II The graph of $y = f(x) - g(x)$ crosses the x -axis exactly three times.
- III $f(x) - g(x)$ is a polynomial of degree at least 3.

Answer: B

I is the definition of an intersection. II fails because the difference could touch the axis rather than cross it at one of the points. III fails because f and g need not be polynomials at all (e.g. $\sin x$ and a line).

Full method - how and why

Intersections and roots. Distinguish a root from a **crossing**, and resist assuming the functions are polynomials.

- 1 I: the graphs meet where $f(x) = g(x)$, so three intersection points means exactly three solutions. **True** - it is the definition.

why: Each intersection point has a distinct x -value, since a graph has one y per x .

- 2 II: at a root of $f - g$ the graph could **touch** the x -axis and turn back rather than cross - a repeated root.

why: Meeting and crossing are different: $y = x^2$ meets the axis but does not cross it.

- 3 So the number of crossings could be fewer than three. **False.**

why: For instance $f - g = x^2(x - 1)$ meets the axis at 0 and 1 but crosses only at 1.

- 4 III: nothing says f and g are polynomials. Take $f(x) = \sin x$ and a suitable straight line g , meeting three times. **False.**

why: Read the hypotheses: only 'the graphs of $y = f(x)$ and $y = g(x)$ ' is given.

- 5 So I only.

why: Match to the list.

Where it goes wrong

- Accepting II by treating 'crosses' as a synonym for 'meets'.
- Accepting III by assuming a degree just because three intersections are mentioned.

Takeaway Intersections = solutions of $f = g$; but crossing the axis needs a sign change, and never assume a function type that was not stated.

Question 6 X2-21 · Graphs of functions · spec MM8.6

$p(x)$ is a polynomial of degree 4 with real coefficients. Which of the following statements must be true?

- I $p(x) = 0$ has at least one real root.
- II $p(x) = 0$ has at most four real roots.
- III The graph of $y = p(x)$ has at least one turning point.

Answer: G

I fails: $x^4 + 1$ has no real roots. II: a degree-4 polynomial has at most 4 roots. III: an even-degree polynomial tends to the same infinity at both ends, so it must turn at least once.

Full method - how and why

Three claims about a quartic. Two follow from degree; the first needs a counterexample.

1 I: must a degree-4 polynomial have a real root? No: $x^4 + 1 \geq 1 > 0$ for all real x . **False.**
why: Even-degree polynomials can stay entirely above the axis - unlike odd-degree ones.

2 II: a polynomial of degree n has at most n real roots, so at most 4. **True.**
why: Each root supplies a linear factor; more than 4 would exceed the degree.

3 III: as $x \rightarrow \pm \infty$, a quartic tends to the same infinity at both ends (the sign of the leading coefficient).
why: Even powers dominate and are positive at both extremes.

4 A continuous function going to the same infinity both ways must turn round somewhere, so it has at least one turning point. **True.**
why: It has a global minimum (or maximum) at a finite point.

5 So II and III.
why: Match to the list.

Where it goes wrong

- Accepting I by analogy with odd-degree polynomials, which **must** have a real root.
- Reading II as 'exactly four roots' - repeated roots and non-real roots reduce the count.

Takeaway Odd degree guarantees a real root; even degree guarantees a turning point. Degree n caps the real roots at n .